

SUMMARY

Recent Ph.D. graduate specializing in Machine Learning and Signal Processing, eager to apply expertise to solve real-world machine learning challenges. Proven experience tackling open problems across diverse multi-modal time series data, alongside a strong foundation in Signal Processing tasks like distribution estimation, filtering, sensing, and change detection in signals.

EDUCATION

Georgia Institute of Technology, Atlanta, GA

Ph.D. in Electrical and Computer Engineering

August 2024

Concentration in Signal Processing and Machine Learning

Thesis: *Detecting and leveraging changes in temporal data*

Devised new methods for detecting distribution shifts, active learning, self-supervised deep learning

Masters in Electrical and Computer Engineering

May 2018

National University of Sciences and Technology, Islamabad, Pakistan

Bachelors in Electrical Engineering

May 2014

SKILLS

Programming/Scripting languages: Python, C/C++, MATLAB, Bash

Machine Learning/Data Science tools: Pytorch, TensorFlow, Keras, Sci-kit learn, SQL, Pandas, SciPY, OpenCV

DevOPS tools: Docker Containers, Ansible, Git, Jenkins, Linux, OpenStack, AzureVMStack

High Performance Computing: OpenMPI

WORK EXPERIENCE

The Coca-Cola Company, Atlanta, GA

Applied Machine Learning Researcher (Fixed term contractual project)

January 2024 - August 2025

- Collaborated with a diverse team of physiologists and biomedical scientists on <https://sweatratecalculator.com>
- Developed novel personalized hydration prediction algorithms via self-supervised learning on wearable multi-modal data.
- Assisting with data trial collection and developed pipelines to process, filter and extract features from raw signals.

Georgia Institute of Technology, Atlanta, GA

Graduate Research Assistant

Aug 2018 - January 2024

- Devised new improved robust change detection methods for real-world noisy settings
- Devised new machine learning methods for deep representation learning, active learning, and domain adaptation

NEC Laboratories America, Princeton, NJ

Research Intern in Machine Learning

May 2021 - Aug 2021

- Devised new self-supervised learning methods for data efficient ordinal classification (labels with inherent order)
- These methods reduce training time by 60 % for ordinal classification
- Developed Signal to Noise ratio classification methods that achieve an accuracy of 95 % for real world optical network data
- Explored how soft-labels are better suited for classifying noisy ordinal data

xFlow Research, Islamabad, Pakistan

Software Engineer

Sep 2014 – June 2016

- Developed a Python benchmarking suite for quantifying cloud computing infrastructure performance
- Tested newly developed Network functions virtualization installation packages on cloud infrastructure for deployment

Ph.D. PROJECTS

Semi-supervised sequence classification through change point detection

- Devised a method to use detected unsupervised change points for obtaining similar dissimilar pairs
- These pairs provide weak supervision that can be used to learn neural network representations in a label efficient manner

WiSAT: An activity recognition system for wheelchair users

- Devised an interpretable classifier for characterizing in-seat movement for paraplegic wheelchair using pressure sensor data
- Assisted in embedding this classifier in iOS/Android applications for end users
- Assisted in devising data collection protocols
- This tracking system is deployed in clinical trials

Sparse metric learning for change point detection

- Devised a new method that uses available change points to improve change point performance
- This method obtained triplet pairs from available change points for learning a metric for Wasserstein distances
- Demonstrated improved change detection performance on human activity, sleep and neuroscience datasets

Collaborative filtering for recommender systems using categorical matrix completion

- Devised a recommendation system that obtained user movie ratings by setting up an ordinal matrix completion problem
- Demonstrated that decomposing a categorical matrix completion problem into 1 bit low-rank matrix completion problems can help improve movie recommendation performance by upto 5 %

Domain adaptation for time series classification via selective channel screening

- Identified scenarios where it is useful to learning invariant feature representations between source and target domains
- Explored appropriate distance metrics in representation space to mine similar class samples across domains
- Designed new method that can ignore spurious and noisy channels that hamper classification performance in new, unlabelled domains. Improves performance over State of the art methods by up to 14 %.

Self-supervision and active learning for learning temporal representations.

- Explored methods for learning self-supervised representations for time series datasets that do not require augmentations
- Devised active label sampling schemes suited for temporal neural network representations for instance classification tasks

TEACHING AND LEADERSHIP EXPERIENCE

Georgia Institute of Technology

Graduate Teaching Assistant

Aug 2020 - May 2021

Conducted office hours. Assisted designing and grading homework problems for the following graduate level courses

- ECE 6270: Convex Optimization
- ECE 6258: Digital Image Processing

SELECTED PUBLICATION AND PREPRINTS

1. **N. Ahad**, M. Davenport, “Semi-supervised Sequence Classification through Change Point Detection”, *AAAI*, 2021
2. **N. Ahad**, E. Dyer, K. Hengen, Y. Xie, M. Davenport, “Learning Sinkhorn divergences for Supervised Change Point Detection”, *In revision, IEEE Transactions on Signal Processing. Preprint* arXiv:2202.04000
3. **N. Ahad**, S. Sonenbom, M. Davenport, S. Sprigle, “Validating a Wheelchair In-Seat Activity Tracker”, *Assistive Technology*, 2021.
4. **N. Ahad**, M. Davenport, Y. Xie, “Data Adaptive Symmetrical CUSUM”, *Sequential Analysis*
5. **N. Ahad***, N. Nadagouda*, E. Dyer, Mark Davenport, “Active Learning for Time Instant Classification”, *Workshop on Data Centeric Machine Learning Research, Int. Conf. on Machine Learning (ICML)*, 2023
6. **N. Ahad**, Mark Davenport, E. Dyer, “Time Series Domain Adaptation via Channel-Selective Representation Alignment”, *to appear in Trans. on Machine Learning Research (TMLR)*, 2025
7. F. Zhu, A. Sedler, H. Grier, **N. Ahad**, M. Davenport, M. Kaufman, A. Giovannucci, C. Pandarinath “Deep inference of latent dynamics with spatio-temporal super-resolution using selective backpropagation through time”, *NeurIPS*, 2021
8. J. Quesada, L. Sathidevi, R. Liu, **N. Ahad**, J. M. Jackson, M. Azabou, J. Xiao, C. Liding, M. J., C. Urzay, W. Gray-Roncal, E. C. Johnson, E. L. Dyer “MTNeuro: A Benchmark for Evaluating Representations of Brain Structure Across Multiple Levels of Abstraction”, *NeurIPS Datasets and Benchmarks Track*, 2022

PATENT APPLICATIONS

- Mizoguchi, L. Tong, Z. Chen, W. Cheng, H. Chen, **N. Ahad**, “Ordinal Classification through Network Decomposition”, US Patent App. 17/896,747, 2023
- L. Tong, T. Mizoguchi, Z. Chen, W. Cheng, H. Chen, and **N. Ahad**, “Semi-supervised framework for efficient time-series ordinal classification”, US Patent App. 18/152,238, 2023